

class13

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##Background Today we will run through a complete RNAseq analysis. The data comes from GEO entry: GSE37704, which is associated with the following publication:

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. “Differential analysis of gene regulation at transcript resolution with RNA-seq”. Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1. Their results and others indicate that HOXA1 is required for lung fibroblast and HeLa cell cycle progression.

Data Import

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metadata <- read.csv("GSE37704_metadata.csv")
counts <- read.csv("GSE37704_featurecounts.csv", row.names=1)

head(colData)
```

```
1 new("standardGeneric", .Data = function (x, ...)
2 standardGeneric("colData"), generic = structure("colData", package = "SummarizedExperiment",
3   package = "SummarizedExperiment", group = list(), valueClass = character(0),
4   signature = "x", default = NULL, skeleton = (function (x,
5     ...))
6   stop(gettextf("invalid call in method dispatch to '%s' (no default method)",
```

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28

ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
colnames(counts)
```

```
[1] "length"      "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370"
[7] "SRR493371"
```

```
head(metadata)
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
counts <- counts[, -1]
test_cols <- !all(colnames(counts) == metadata$id)
```

Also lets remove low count genes

```
tot.counts <- rowSums(counts)
head(tot.counts)
```

ENSG00000186092	ENSG00000279928	ENSG00000279457	ENSG00000278566	ENSG00000273547
0	0	183	0	0
ENSG00000187634				
1129				

Let's remove all zero count genes

```
zero.inds <- tot.counts==0
head(zero.inds)
```

ENSG00000186092	ENSG00000279928	ENSG00000279457	ENSG00000278566	ENSG00000273547
TRUE	TRUE	FALSE	TRUE	TRUE
ENSG00000187634				
FALSE				

```
counts <- counts[!zero.inds,]
```

```
if(test_cols) {
  message("WOW... there us a problemm with the metadata counts setup")
}
```

```
!all(c(T,T,T,F))
```

```
[1] TRUE
```

Setup for Deseq

```
library(DESeq2)
```

```
head(counts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
dds <- DESeqDataSetFromMatrix(countData=counts,
                              colData = metadata,
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run Deseq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get Results

```
res <- results(dds)
```

```
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01

```

ENSG00000187634 183.2296      0.4264571 0.1402658    3.040350 2.36304e-03
ENSG00000188976 1651.1881    -0.6927205 0.0548465   -12.630156 1.43993e-36
ENSG00000187961 209.6379      0.7297556 0.1318599    5.534326 3.12428e-08
ENSG00000187583 47.2551       0.0405765 0.2718928    0.149237 8.81366e-01
ENSG00000187642 11.9798       0.5428105 0.5215598    1.040744 2.97994e-01
      padj
      <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01

```

Add Anotation

```

library("AnnotationDbi")
library("org.Hs.eg.db")

```

```
columns(org.Hs.eg.db)
```

```

[1] "ACCNUM"      "ALIAS"       "ENSEMBL"     "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"      "EVIDENCE"    "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"          "GOALL"       "IPI"          "MAP"
[16] "OMIM"        "ONTOLOGY"    "ONTOLOGYALL" "PATH"         "PFAM"
[21] "PMID"        "PROSITE"     "REFSEQ"      "SYMBOL"       "UCSCKG"
[26] "UNIPROT"

```

```

res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",)

```

'select()' returned 1:many mapping between keys and columns


```
res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 8 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez		
	<numeric>	<character>	<character>		
ENSG00000279457	6.86555e-01	NA	NA		
ENSG00000187634	5.15718e-03	SAMD11	148398		
ENSG00000188976	1.76553e-35	NOC2L	26155		
ENSG00000187961	1.13413e-07	KLHL17	339451		
ENSG00000187583	9.19031e-01	PLEKHN1	84069		
ENSG00000187642	4.03379e-01	PERM1	84808		
ENSG00000188290	1.30538e-24	HES4	57801		
ENSG00000187608	2.37452e-02	ISG15	9636		
ENSG00000188157	4.21970e-16	AGRN	375790		
ENSG00000237330	NA	RNF223	401934		

Visualize Results

```
library(ggplot2)

my_cols <- rep("gray", nrow(res))
my_cols[abs(res$log2FoldChange)>=2] <- "blue"
my_cols
```

```
[1] "gray" "gray" "gray" "gray" "gray" "gray" "blue" "gray" "gray" "gray"
[11] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[21] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[31] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[41] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[51] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[61] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[71] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "blue"
[81] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[91] "gray" "gray" "gray" "gray" "blue" "gray" "gray" "gray" "gray" "gray"
[101] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[111] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[121] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[131] "gray" "gray" "gray" "gray" "gray" "blue" "gray" "gray" "gray" "blue"
[141] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[151] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[161] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[171] "gray" "gray" "gray" "blue" "gray" "gray" "gray" "gray" "gray" "gray"
[181] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[191] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[201] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[211] "gray" "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[221] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[231] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "blue"
[241] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[251] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "blue" "gray" "gray"
[261] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "blue" "gray"
[271] "gray" "gray" "blue" "gray" "gray" "blue" "gray" "gray" "gray" "gray"
[281] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[291] "gray" "gray" "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[301] "gray" "gray" "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[311] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[321] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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```

[15811] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15821] "gray" "gray" "gray" "gray" "gray" "gray" "blue" "gray" "gray" "gray"
[15831] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15841] "gray" "gray" "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15851] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15861] "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15871] "gray" "gray" "gray" "gray" "gray" "gray" "blue" "gray" "gray" "gray"
[15881] "gray" "blue" "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15891] "gray" "blue" "blue" "blue" "gray" "gray" "gray" "gray" "gray" "gray"
[15901] "gray" "gray" "gray" "blue" "blue" "blue" "gray" "gray" "gray" "gray"
[15911] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15921] "gray" "gray" "gray" "blue" "gray" "blue" "gray" "gray" "blue" "gray"
[15931] "blue" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15941] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15951] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15961] "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray" "gray"
[15971] "gray" "gray" "gray" "gray" "gray"

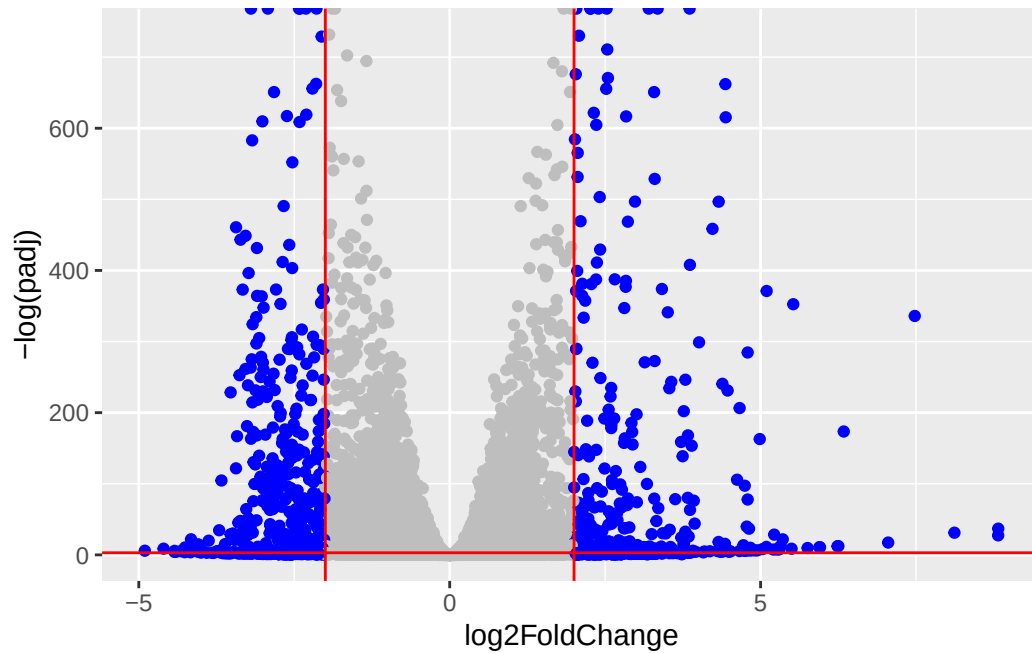
```

```

ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(col=my_cols) +
  geom_vline(xintercept=c(-2,2), col="red") +
  geom_hline(yintercept=-log(0.05), col="red")

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Pathway Analysis

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)
# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"    "1544"  "1548"  "1549"  "1553"  "7498"  "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```



```

[1] "10"      "1066"    "10720"   "10941"   "151531"  "1548"    "1549"    "1551"
[9] "1553"    "1576"    "1577"    "1806"    "1807"    "1890"    "221223"  "2990"
[17] "3251"    "3614"    "3615"    "3704"    "51733"   "54490"   "54575"   "54576"
[25] "54577"   "54578"   "54579"   "54600"   "54657"   "54658"   "54659"   "54963"
[33] "574537"  "64816"   "7083"    "7084"    "7172"    "7363"    "7364"    "7365"
[41] "7366"    "7367"    "7371"    "7372"    "7378"    "7498"    "79799"   "83549"
[49] "8824"    "8833"    "9"       "978"

```

```
$`hsa00230 Purine metabolism`
```

```

[1] "100"      "10201"   "10606"   "10621"   "10622"   "10623"   "107"     "10714"
[9] "108"      "10846"   "109"     "111"     "11128"   "11164"   "112"     "113"
[17] "114"      "115"     "122481"  "122622"  "124583"  "132"     "158"     "159"
[25] "1633"     "171568"  "1716"    "196883"  "203"     "204"     "205"     "221823"
[33] "2272"     "22978"   "23649"   "246721"  "25885"   "2618"    "26289"   "270"
[41] "271"     "27115"   "272"     "2766"    "2977"    "2982"    "2983"    "2984"
[49] "2986"     "2987"   "29922"   "3000"    "30833"   "30834"   "318"     "3251"
[57] "353"     "3614"    "3615"    "3704"    "377841"  "471"     "4830"    "4831"
[65] "4832"     "4833"   "4860"    "4881"    "4882"    "4907"    "50484"   "50940"
[73] "51082"    "51251"   "51292"   "5136"    "5137"    "5138"    "5139"    "5140"
[81] "5141"     "5142"   "5143"    "5144"    "5145"    "5146"    "5147"    "5148"
[89] "5149"     "5150"   "5151"    "5152"    "5153"    "5158"    "5167"    "5169"
[97] "51728"    "5198"   "5236"    "5313"    "5315"    "53343"   "54107"   "5422"
[105] "5424"     "5425"   "5426"    "5427"    "5430"    "5431"    "5432"    "5433"
[113] "5434"     "5435"   "5436"    "5437"    "5438"    "5439"    "5440"    "5441"
[121] "5471"     "548644" "55276"   "5557"    "5558"    "55703"   "55811"   "55821"
[129] "5631"     "5634"   "56655"   "56953"   "56985"   "57804"   "58497"   "6240"
[137] "6241"     "64425"   "646625"  "654364"  "661"     "7498"    "8382"    "84172"
[145] "84265"    "84284"   "84618"   "8622"    "8654"    "87178"   "8833"    "9060"
[153] "9061"     "93034"   "953"     "9533"    "954"     "955"     "956"     "957"
[161] "9583"     "9615"

```

```

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)

```

```

      <NA>      148398      26155      339451      84069      84808
0.17925708 0.42645712 -0.69272046 0.72975561 0.04057653 0.54281049

```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names  
[1] "greater" "less" "stats"
```

```
## $names  
## [1] "greater" "less" "stats"
```

```
# Look at the first few down (less) pathways  
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.246882e-03	-3.059466	1.246882e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.066915975	144	1.246882e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

```
The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG  
license agreement (details at http://www.kegg.jp/kegg/legal.html).
```

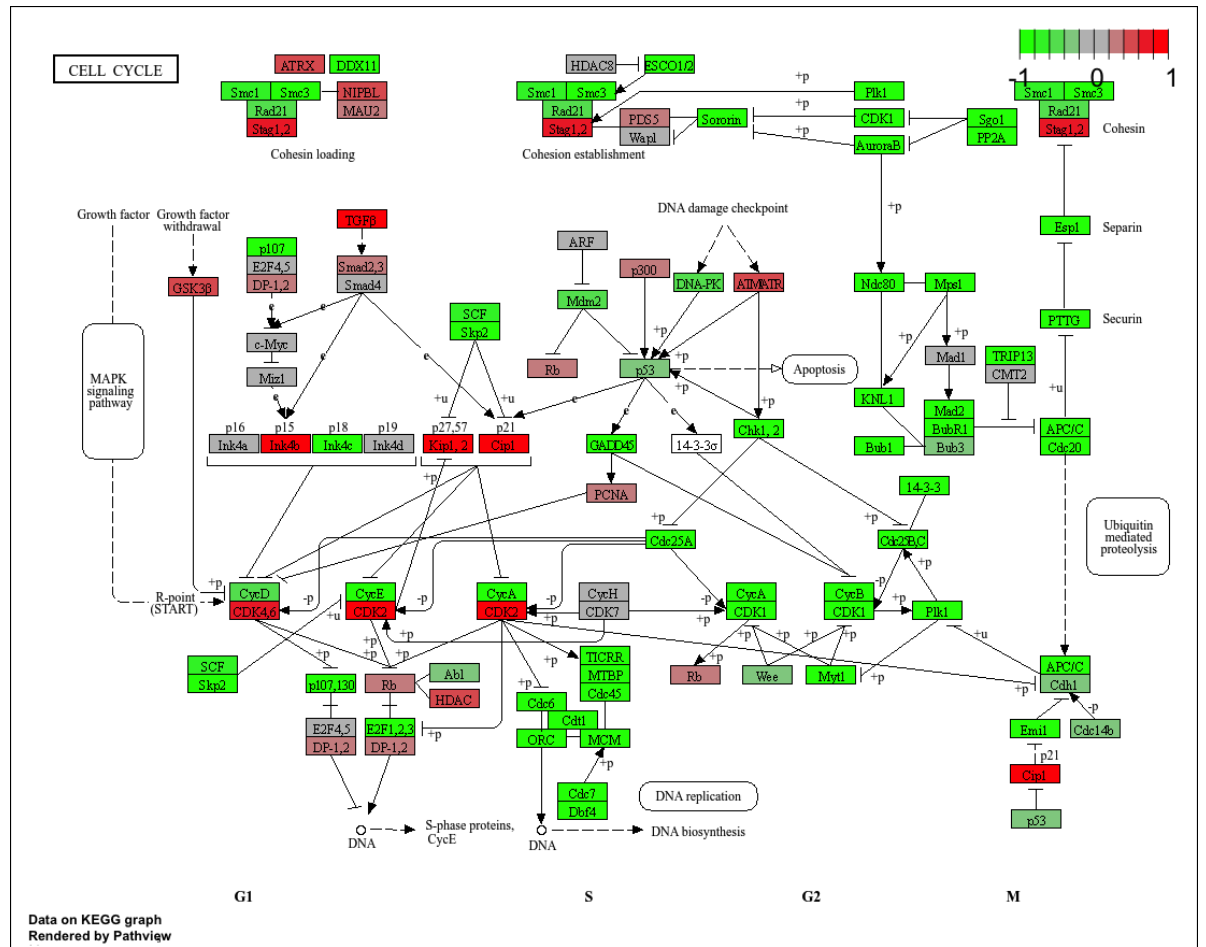
```
#####
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns
```

Info: Working in directory /Users/ebruy/Desktop/UCSD/Bioinformatics 213/class1

Info: Writing image file hsa04110.pathview.png



Let's try GO analysis and compare to KEGG analysis

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```
head(gobpres$less)
```

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

Reactome

Some folks really like Reactome online (i.e. their webpage viewer < <https://reactome.org/PathwayBrowser/#TOOL=AT>) rather than the R package of the same name (available from bioconductor).

To use the website viewer we want to upload our set of gene symbols for the genes we want to focus on (here those with a P-value below 0.05)

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]

write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Save Results

```
write.csv(res,file="my_results.csv")
```

```
save(res,counts,metadata,file="my_results.RData")
```